

Safety Data Sheet

Caprolactam extract for storage

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(30043509/SDS_GEN_US/EN)

1. Identification

Product identifier used on the label

Caprolactam extract for storage

Recommended use of the chemical and restriction on use

Recommended use*: Chemical

Recommended use*: for the production of homopolymerisates and copolymerisates; initial product for chemical syntheses

Unsuitable for use: Not intended for sale to or use by the general public.

* The "Recommended use" identified for this product is provided solely to comply with a Federal requirement and is not part of the seller's published specification. The terms of this Safety Data Sheet (SDS) do not create or infer any warranty, express or implied, including by incorporation into or reference in the seller's sales agreement.

Details of the supplier of the safety data sheet

Company:

BASF CORPORATION
100 Park Avenue
Florham Park, NJ 07932, USA

Telephone: +1 973 245-6000

Emergency telephone number

24 Hour Emergency Response Information

CHEMTREC: 1-800-424-9300

BASF HOTLINE: 1-800-832-HELP (4357)

Other means of identification

Chemical family: organic compounds

2. Hazards Identification

According to Regulation 2024 OSHA Hazard Communication Standard; 29 CFR Part 1910.1200

Classification of the product

Acute Tox.	4 (Inhalation - dust)	Acute toxicity
Acute Tox.	4 (oral)	Acute toxicity
Skin Irrit.	2	Skin irritation
Eye Irrit.	2A	Eye irritation

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STOT SE

3 (irritating to
respiratory system)

Specific target organ toxicity — single exposure

Label elements

Pictogram:



Signal Word:
Warning

Hazard Statement:

H319	Causes serious eye irritation.
H315	Causes skin irritation.
H335	May cause respiratory irritation.
H302 + H332	Harmful if swallowed or if inhaled.

Precautionary Statements (Prevention):

P271	Use only outdoors or in a well-ventilated area.
P280	Wear protective gloves and eye protection or face protection.
P261	Avoid breathing mist or vapour or spray.
P270	Do not eat, drink or smoke when using this product.
P264	Wash contaminated body parts thoroughly after handling.

Precautionary Statements (Response):

P312	Call a POISON CENTER or physician if you feel unwell.
P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P304 + P340	IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P302 + P352	IF ON SKIN: Wash with plenty of soap and water.
P330	Rinse mouth.
P332 + P313	If skin irritation occurs: Get medical attention.
P337 + P313	If eye irritation persists: Get medical attention.
P362 + P364	Take off contaminated clothing and wash it before reuse.

Precautionary Statements (Storage):

P403 + P233	Store in a well-ventilated place. Keep container tightly closed.
P405	Store locked up.

Precautionary Statements (Disposal):

P501	Dispose of contents/container in accordance with local regulations.
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Hazards not otherwise classified

No specific dangers known, if the regulations/notes for storage and handling are considered.

3. Composition / Information on Ingredients

According to Regulation 2024 OSHA Hazard Communication Standard; 29 CFR Part 1910.1200

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caprolactam

CAS Number: 105-60-2
Content (W/W): > 65.0 - < 85.0%
Synonym: Caprolactam

The actual concentration is withheld as a trade secret.

4. First-Aid Measures

Description of first aid measures

General advice:

Immediately remove contaminated clothing. Avoid contact with the skin, eyes and clothing.

If inhaled:

Keep patient calm, remove to fresh air, seek medical attention. Assist in breathing if necessary.

If on skin:

Wash thoroughly with soap and water

If in eyes:

Hold eyes open and rinse slowly and gently with water for 15 to 20 minutes. Remove contact lenses, if present, after first 5 minutes, then continue rinsing. Seek medical attention.

If swallowed:

Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.

Information on: caprolactam

Symptoms: Overexposure may cause: , headache, tachycardia, hypertension, hypotension (low blood pressure), fever, anorexia, epigastric distress, nausea, convulsions

Hazards: No applicable information available.

Indication of any immediate medical attention and special treatment needed

Note to physician

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Extinguishing media

Suitable extinguishing media:
foam, carbon dioxide, water spray

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Unsuitable extinguishing media for safety reasons:
water jet

Special hazards arising from the substance or mixture

Hazards during fire-fighting:

hydrogen cyanide, nitrogen oxides

The substances/groups of substances mentioned can be released in case of fire.

Advice for fire-fighters

Protective equipment for fire-fighting:

Firefighters should be equipped with self-contained breathing apparatus and turn-out gear.

Further information:

Product itself is non-combustible; fire extinguishing method of surrounding areas must be considered. Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems.

6. Accidental release measures

Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Avoid contact with skin and eyes. Use breathing apparatus if exposed to vapours/dust/aerosol. Information regarding personal protective measures, see section 8.

Environmental precautions

Do not empty into drains. Retain and dispose of contaminated wash water.

Methods and material for containment and cleaning up

For large amounts: Pump off product.

For residues: Pick up with suitable absorbent material (e.g. sand, sawdust, general-purpose binder, kieselguhr). Rinse away with water.

7. Handling and Storage

Precautions for safe handling

Ensure thorough ventilation of stores and work areas. Avoid contact with skin and eyes. Wear suitable protective clothing and eye/face protection. Handle in accordance with good industrial hygiene and safety practice.

Protection against fire and explosion:

No special precautions necessary.

Conditions for safe storage, including any incompatibilities

Segregate from acids and bases. Segregate from oxidants.

Suitable materials for containers: Stainless steel 1.4301 (V2), Stainless steel 1.4401, Carbon steel (Iron), High density polyethylene (HDPE)

8. Exposure Controls/Personal Protection

Components with occupational exposure limits

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caprolactam	NIOSH, US:	REL value 0.22 ppm 1 mg/m3 vapour ;
	OSHA Z1A:	STEL value 3 mg/m3 dust ;
	OSHA Z1A:	STEL value 10 ppm 40 mg/m3 vapour ;
	NIOSH, US:	STEL value 0.66 ppm 3 mg/m3 vapour ;
	OSHA Z1A:	TWA value 1 mg/m3 dust ;
	OSHA Z1A:	TWA value 5 ppm 20 mg/m3 vapour ;
	NIOSH, US:	REL value 1 mg/m3 dust ;
	NIOSH, US:	STEL value 3 mg/m3 dust ;
	ACGIH, US:	TWA value 5 mg/m3 Inhalable fraction and vapor ;

Advice on system design:

Ensure adequate ventilation. Provide local exhaust ventilation to maintain recommended P.E.L.

Personal protective equipment

Respiratory protection:

Wear a NIOSH-certified (or equivalent) organic vapour/particulate respirator. Do not exceed the maximum use concentration for the respirator facepiece/cartridge combination. For emergency or non-routine, high exposure situations, use a NIOSH-certified full facepiece pressure demand self-contained breathing apparatus (SCBA) or a full facepiece pressure demand supplied-air respirator (SAR) with escape provisions.

Observe OSHA regulations for respirator use (29 CFR 1910.134).

Hand protection:

Chemical resistant protective gloves should be worn to prevent all skin contact., Suitable materials may include, nitrile rubber (Buna N), chloroprene rubber (Neoprene), polyvinylchloride (Pylox), Consult with glove manufacturer for testing data., Protective glove selection must be based on the user's assessment of the workplace hazards.

Eye protection:

Tightly fitting safety goggles (chemical goggles). Wear face shield if splashing hazard exists.

Body protection:

Body protection must be chosen depending on activity and possible exposure, e.g. head protection, apron, protective boots, chemical-protection suit.

General safety and hygiene measures:

Handle in accordance with good industrial hygiene and safety practice. Take off immediately all contaminated clothing. At the end of the shift the skin should be cleaned and skin-care agents applied.

9. Physical and Chemical Properties

Physical state:	liquid	
Form:	solution	
Odour:	pungent odour	
Odour threshold:	No data available.	
Colour:	slightly yellow	
pH value:	8	(pH Meter)

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solidification temperature:	14.5 °C (80 %(m)) 41.4 °C (90 %(m))	(other)
Melting point:	No data available.	
Freezing point:	No data available.	
Boiling range:	No data available.	
<i>Information on: Water</i>		
Boiling point:	100 °C	

Sublimation point:	No applicable information available.	
Flash point:	not applicable	
Flammability:	not highly flammable	(other)
Lower explosion limit:	not applicable	
Upper explosion limit:	not applicable	
Autoignition:	not applicable	
Vapour pressure:	No data available.	
<i>Information on: Caprolactam</i>		
Vapour pressure:	0.0013 hPa (20 °C) Literature data. 0.089 hPa (60 °C) Literature data.	

Density:	1 g/cm3 (20 °C)	(OECD Guideline 109)
Relative density:	No data available.	
Relative vapour density:	not determined	
Partitioning coefficient n-octanol/water (log Pow):	No data available.	
<i>Information on: caprolactam</i>		
Partitioning coefficient n-octanol/water (log Pow):	0.12 (25 °C)	(OECD Guideline 107)

Self-ignition temperature:	Based on its structural properties the product is not classified as self-igniting.	
Thermal decomposition:	No decomposition if correctly stored and handled.	
Viscosity, dynamic:	not determined	
Viscosity, kinematic:	not determined	
Solubility in water:	miscible	
Solubility (quantitative):	No data available.	
Solubility (qualitative):	No data available.	
Molecular weight:	113.16 g/mol	
Evaporation rate:	Value can be approximated from Henry's Law Constant or vapor pressure.	

Particle characteristics

Particle size distribution: The substance / product is marketed or used in a non solid or granular form.

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10. Stability and Reactivity

Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals:

No corrosive effect on metal.

Oxidizing properties:

Based on its structural properties the product is not classified as oxidizing.

Chemical stability

The product is stable if stored and handled as prescribed/indicated.

Possibility of hazardous reactions

Reacts with oxidizing agents. Polymerization coupled with heat formation.

Conditions to avoid

Temperature: > 100 degrees Celsius

Avoid all sources of ignition: heat, sparks, open flame. Avoid formation of polymers in valves and pipes.

Incompatible materials

strong acids, bases

oxidizing agents

Hazardous decomposition products

Decomposition products:

Thermal decomposition products: No hazardous decomposition products if stored and handled as prescribed/indicated., Incomplete combustion results in formation of toxic gases, containing mainly carbon monoxide and carbon dioxide.

Thermal decomposition:

No decomposition if correctly stored and handled.

11. Toxicological information

Primary routes of exposure

Routes of entry for solids and liquids are ingestion and inhalation, but may include eye or skin contact. Routes of entry for gases include inhalation and eye contact. Skin contact may be a route of entry for liquefied gases.

Acute Toxicity/Effects

Acute toxicity

Assessment of acute toxicity: Harmful by inhalation and if swallowed.

Oral

Type of value: ATE

Value: 1,560 mg/kg

Information on: caprolactam

Type of value: LD50

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Species: rat (female)

Value: 1,475 mg/kg (Directive 84/449/EEC, B.1)

Inhalation

Type of value: ATE

Value: > 20.0000 mg/l

Determined for vapor

Type of value: ATE

Value: > 5.0000 mg/l

Determined for mist

Information on: caprolactam

Type of value: LC50

Species: rat (male/female)

Value: approx. 8.16 mg/l (BASF-Test)

Exposure time: 4 h

An aerosol with respirable particles was tested.

Dermal

Type of value: ATE

Value: > 5,000 mg/kg

Information on: caprolactam

Type of value: LD50

Species: rat (male/female)

Value: > 2,000 mg/kg (Directive 92/69/EEC, B.3)

Assessment other acute effects

Assessment of STOT single:

Causes temporary irritation of the respiratory tract.

Irritation / corrosion

Assessment of irritating effects: Causes serious eye irritation. Causes skin irritation.

Information on: caprolactam

Assessment of irritating effects: May cause slight irritation to the skin. May cause slight irritation to the eyes.

Sensitization

Assessment of sensitization: Based on available data, the classification criteria are not met.

Information on: caprolactam

modified Buehler test

Species: guinea pig

Result: Non-sensitizing.

Method: OECD Guideline 406

Aspiration Hazard

No aspiration hazard expected.

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Chronic Toxicity/Effects

Repeated dose toxicity

Assessment of repeated dose toxicity: Based on available data, the classification criteria are not met.

Information on: caprolactam

Assessment of repeated dose toxicity: After repeated exposure the prominent effect is local irritation. The substance may cause damage to the upper respiratory tract after repeated inhalation, as shown in animal studies.

Genetic toxicity

Assessment of mutagenicity: Based on available data, the classification criteria are not met.

Carcinogenicity

Assessment of carcinogenicity: Based on available data, the classification criteria are not met.

Reproductive toxicity

Assessment of reproduction toxicity: Based on available data, the classification criteria are not met.

Teratogenicity

Assessment of teratogenicity: Based on available data, the classification criteria are not met.

Other Information

The product has not been tested. The statements on toxicology have been derived from the properties of the individual components.

12. Ecological Information

Toxicity

Aquatic toxicity

Assessment of aquatic toxicity:

Based on available data, the classification criteria are not met.

Aquatic toxicity

Information on: caprolactam

Assessment of aquatic toxicity:

There is a high probability that the product is not acutely harmful to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Toxicity to fish

Information on: caprolactam

LC0 (96 h) 100 mg/l, Oryzias latipes (OECD Guideline 203, semistatic)

LC50 (96 h) 500 - 1,000 mg/l, Salmo gairdneri, syn. O. mykiss (OECD 203; ISO 7346; 84/449/EWG, C.1, static)

LC50 (96 h) 707.1 mg/l, Salmo gairdneri, syn. O. mykiss (OECD 203; ISO 7346; 84/449/EWG, C.1, static)

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Aquatic invertebrates

Information on: caprolactam

EC50 (48 h) > 1,000 mg/l, *Daphnia magna* (OECD Guideline 202, part 1, static)

EC50 (48 h) > 500 mg/l, *Daphnia magna* (DIN 38412 Part 11, static)

EC50 (48 h) > 500 mg/l, *Daphnia magna* (DIN 38412 Part 11, static)

Aquatic plants

Information on: caprolactam

No observed effect concentration (72 h) 1,000 mg/l (growth rate), *Pseudokirchneriella subcapitata* (OECD Guideline 201, static)

EC50 (72 h) > 1,000 mg/l (growth rate), *Selenastrum capricornutum* (OECD Guideline 201, static)

EC50 (72 h) 427.5 mg/l (growth rate), *Scenedesmus subspicatus* (DIN 38412 Part 9, static)

EC50 (72 h) > 1,000 mg/l (biomass), *Selenastrum capricornutum* (OECD Guideline 201, static)

Chronic toxicity to fish

Information on: caprolactam

Study scientifically not justified.

Chronic toxicity to aquatic invertebrates

Information on: caprolactam

No observed effect concentration (21 d) 100 mg/l, *Daphnia magna* (OECD Guideline 211, semistatic)

Assessment of terrestrial toxicity

Information on: caprolactam

No data available.

Study scientifically not justified.

Microorganisms/Effect on activated sludge

Toxicity to microorganisms

Information on: caprolactam

other aquatic

bacterium/EC50 (17 h): 4,240 mg/l

Persistence and degradability

Assessment biodegradation and elimination (H2O)

Information on: caprolactam

Readily biodegradable (according to OECD criteria).

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Elimination information

Information on: caprolactam

82 % BOD of the ThOD (14 d) (OECD 301C; ISO 9408; 92/69/EWG, C.4-F) (aerobic, activated sludge) Readily biodegradable (according to OECD criteria).

Assessment of stability in water

Information on: caprolactam

In contact with water the substance will hydrolyse slowly.

Bioaccumulative potential

Bioaccumulation potential

Information on: caprolactam

Because of the *n*-octanol/water distribution coefficient ($\log P_{ow}$) accumulation in organisms is not to be expected.

Mobility in soil

Assessment transport between environmental compartments

Information on: caprolactam

The substance will not evaporate into the atmosphere from the water surface.
Adsorption to solid soil phase is not expected.

Additional information

Sum parameter

Information on: Caprolactam

Chemical oxygen demand (COD): 1,960 mg/g

Information on: Caprolactam

Biochemical oxygen demand (BOD): 1,110 mg/g

Other ecotoxicological advice:
Do not release untreated into natural waters.

13. Disposal considerations

Waste disposal of substance:

Dispose of in accordance with national, state and local regulations.

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Container disposal:

Uncleaned empties should be disposed of in the same manner as the contents.

14. Transport Information

Land transport

USDOT

Not classified as a dangerous good under transport regulations

Sea transport

IMDG

Not classified as a dangerous good under transport regulations

Air transport

IATA/ICAO

Not classified as a dangerous good under transport regulations

15. Regulatory Information

Federal Regulations

Registration status:

Chemical TSCA, US

All substances are TSCA listed and active.

EPCRA 311/312 (Hazard categories): Refer to SDS section 2 for GHS hazard classes applicable for this product.

State regulations

State RTK

PA

NJ

CAS Number

105-60-2

105-60-2

Chemical name

caprolactam

caprolactam

NFPA Hazard codes:

Health: 2

Fire: 0

Reactivity: 0

Special:

16. Other Information

SDS Prepared by:

BASF NA Product Regulations

SDS Prepared on: 2025/11/07

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operations on society and the environment during production, storage, transport, use and disposal of our products.

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